SRAF-Net: A Scene-Relevant Anchor-Free Object Detection Network in Remote Sensing Images

Junmin Liu, *Member, IEEE*, Shijie Li, Changsheng Zhou, Xiangyong Cao, Yong Gao, and Bo Wang

***Abstract*— Object detection is a fundamental and important task in the analysis of *remote sensing images* (RSIs), and existing deep learning-based object detection models in this literature strongly rely on predefined anchor boxes and encounter redesigned difficulties related to anchors. In addition, they often ignore the scene-contextual information that objects are usually closely related to their surrounding scene. To deal with these problems, we propose an anchor-free network, referred to as *scene-relevant anchor-free network* (SRAF-Net), for object detection in RSIs. The SRAF-Net first captures the scene-contextual features of objects by using a designed *scene-enhanced feature pyramid network* (SE-FPN) and then performs more accurate detection by implementing a *scene auxiliary detection head* (SADH), which can predict the existence of the objects with the help of the scene-contextual features extracted from the SE-FPN. To deal with insufficient scene diversity in the training stage, a simple yet effective data augmentation module, termed *balanced mixup data augment* (BMDA), is introduced by linearly expanding the training dataset to improve the generalization of SRAF-Net. Comprehensive experiments on three publicly available challenging remote sensing datasets demonstrate the effectiveness of the proposed method. The codes will be made publicly available at https://github.com/Complicateddd/SRAF-Net.**

***Index Terms*— Anchor-free, object detection, remote sensing, scene-contextual information.**

I. Introduction

THE rapid development of remote sensing technologies has significantly increased the quality and quantity of *remote sensing images* (RSIs), e.g., for greatly capturing spatial–spectral features, graph convolutional networks [1] and multimodal deep learning [2] have recently been used in *hyperspectral* image classification. With these advanced innovations and the recent success in hyperspectral imagery [3]–[5], remote sensing-related technologies have been greatly promoted. However, the large scale and complexity of RSIs lead to their manual analysis impractical or even impossible. Therefore, automatic analysis of RSIs has become a primary concern in recent years. *Object detection*, which seeks to recognize and locate object instances of certain categories in images, is one of the most essential tasks in remotely sensed image understanding, and has a wide range of applications, such as intelligent monitoring [6], urban planning [7], object tracking [8], urban traffic management [9], and drone exploration tasks [10].

More recently, techniques based on *deep learning* have significantly boosted the *state-of-the-art* (SOTA) approaches for object detection. Representative works are those based on region proposals (e.g., R-CNN [11], Fast R-CNN [12], Faster R-CNN [13], and Cascade R-CNN [15]), *single shot multibox detector* (SSD) [16], *you only look once* (YOLO) [17], and so on. However, these approaches are usually designed for *ground-based scene images*. To narrow the performance gap between object detection in ground-based scene images and RSIs, a *rotation-invariant convolutional neural network* (RICNN) detection model in an R-CNN framework was proposed in [18]. Furthermore, a *position-sensitive balancing* framework was proposed in [19] to improve the detection accuracy and speed by adding a position-sensitive pooling layer to the fully convolutional *residual network* (ResNet) [20]. Among these achievements, the anchor mechanism plays a crucial role. Specifically, the anchors in modern detectors are regression references and classification candidates to search proposals for two-stage detectors [13], [15], [21] or final bounding boxes for one-stage detectors [16], [19]. However, the predefined anchors require many handcrafted hyperparameters, such as the *number of anchors*, their *scales*, and *aspect ratios*; thus, the performance and generalization ability of these anchor-based detectors are limited by the design of anchors.

To ease the burden of anchor design and strive for simplicity of detection models, studies have focused on discarding the anchor mechanism and developing anchor-free detection models. These anchor-free models directly find objects without predefined anchors in two different ways. One is to first locate key points and then regress the sizes and offset of objects. Representative works in this way are Cornernet [22] and CenterNet [23]. Another is to use the center area of objects to define positives and then predict the four distances from positives to the object boundary. Studies belonging to this way are Foveabox [24], FCOS [25], and DenseBox [26]. These anchor-free models have eliminated the handcrafted hyperparameters associated with anchors and achieved comparable and

Manuscript received June 8, 2021; revised September 19, 2021; accepted October 25, 2021. Date of publication November 2, 2021; date of current version February 14, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61877049 and Grant 61906151. *(Corresponding author: Junmin Liu.)*

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Digital Object Identifier 10.1109/TGRS.2021.3124959

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even better performances, showing more potential in terms of detection accuracy and generalization. With the development of the anchor-free method in the field of RSIs, scholars have obtained some achievements, e.g., DPAFPN [27] designed a dense path aggregation feature pyramid network (FPN) to avoid information loss during shallow feature transfer. Based on CenterNet, CenterFPANet [28] introduced an FPA module attention mechanism to solve the difficulty caused by the large size range of remote sensing objects. Other works [29]–[31] have achieved gratifying results, but they are task-special or class-special. In this work, we propose a more universal anchor-free object detector in RSIs.

In fact, objects have a strong correlation with the *contextual information* of their scenes, i.e., object instances for a specific category appear, with a very high probability, in certain scenes. Taking this into account, some studies exploiting the contextual information (or the scene prior) for object detection have gradually been proposed and show that contextual information may be helpful to improve the performance of object detection in both the ground-based scene images and RSIs. For example, Zhang *et al.* [32] proposed a *context-aware detection network* by fusing contextual information and an attention mechanism into the classical anchor-based model (e.g., Faster R-CNN with FPN [13]). Tao *et al.* [33] added a scene classification module to incorporate the scene prior for vehicle detection. Despite the achievements in the exploitation of scene information, these works are task-specific, and most of them are anchor-based models while not specifically designed for anchor-free models.

To the best of our knowledge, there is not an anchor-free universal object detection method that integrates the scene-relevant information in the field of remote sensing. The motivation of this work is to utilize the merits of an anchor-free framework, and also, considering the contextual information provided by the scene, we propose a novel *scene-relevant anchor-free network* (SRAF-Net) for object detection in RSIs. Specifically, the SRAF-Net consists of a *scene-enhanced FPN* (SE-FPN) that takes FPN [14] as a backbone assisted by a *scene refine module* (SRM) to extract scene-level features. The SE-FPN is very helpful for capturing the scene-level semantic features. The contextual features extracted by the SE-FPN enabled us to design a *scene auxiliary detection head* (SADH) module in the SRAF-Net for predicting the probabilities of specific objects that appeared in specific scenes. Therefore, we significantly improve detection accuracy by combining the best of both SE-FPN and SADH. Moreover, we develop a simple yet effective data augmentation technique to train the proposed SRAF-Net. To summarize, our contributions are given as follows.

1) We propose SRAF-Net, instead of task-special or class-special, a more general multiclass anchor-free framework for multiclass object detection in RSIs.
2) We consider scene-relevant information and design a scene-enhanced multiscale feature extraction network to capture scene-contextual features to enhance the discriminative abilities of SRAF-Net.
3) We design a novel prediction head to perform classification of objects based on the scene-contextual features, and the advantages of anchor-free and the utilization of scene-contextual information are wisely integrated.
4) To deal with the challenges of model parameter optimization, a straightforward data augmentation technique is introduced by carefully mixing different scenes. Using this technique, SRAF-Net shows great robustness and generalization ability to complex backgrounds in RSIs.

The remainder of this article is structured as follows. Section II reviews previous studies related to this work. The proposed SRAF-Net is presented in Section III. Section IV demonstrates the effectiveness of SRAF-Net and provides a comparison result with the numerous representative methods on three challenging RSI datasets. It also includes an ablation study on our SRAF-Net. Section V concludes this work.

II. Related Work

A. Anchor-Free Methods

Most SOTA detectors rely on anchor boxes [13], [15], [16], [34], which are regression references and classification candidates to predict object locations and categories, and are widely used as a common component (e.g., region proposal network [13]) in object detection frameworks. Although the anchors, by enumerating possible locations, are helpful in reducing the searching space and improving the detection accuracy, their handcrafted design, including factors such as the number of anchors, their sizes, and their aspect ratios, also has some drawbacks [24] that limit the performance and generalization ability of current detection frameworks. Therefore, most recent studies have focused on giving up the anchor mechanism and trying to develop frameworks without candidate anchor boxes, i.e., anchor-free methods.

One of the most famous anchor-free methods is YOLOv1 [17], where the parameters of a bounding box are regressed directly without any anchor. Another successful detection framework is DenseBox [26], which utilizes every pixel of the feature map to regress the four-side distances between the candidate center point and the four bounds of the object candidate containing it. To further boost the localization accuracy of the anchor-free detectors, UnitBox [35] was proposed by directly using *intersection over union* (IoU) as the localization loss function. Unlike DenseBox and UnitBox, FCOS [25] attempts to locate the four-side distances by taking advantage of all points in a ground-truth bounding box. After years of effort, a series of anchor-free methods [36], [37] have been developed and shown to provide a comparable recall with anchor-based methods. Compared to FCOS, CornerNet [22] predicts the upper left corner and lower right corner of the ground-truth box. Instead of predicting two corner points, CenterNet [23] tries to improve the localization accuracy by predicting more key points of the objects. Furthermore, anchor-free methods have been gradually studied in RSIs. For example, Gu *et al.* [28] proposed CenterFPA that introduces the attention mechanism to dynamically adjust the weights of each level of feature maps. Li *et al.* [27] introduced an anchor-free single-stage detector with a designed dense path aggregation FPN (DPAFPN). Shi *et al.* [30] used central points, orientations, scales, and offsets of objects to

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detect vehicles. Zhang *et al.* [29] proposed a Gaussian mask to model the ship based on its geometry characteristics and solved the mislabeling problem of rotation bounding boxes. Sun *et al.* [31] proposed a multitask detection head to regress irregular garbage dumps in a more accurate and efficient way. However, the above methods are task-special or class-special; there is still some room for the anchor-free framework applied for object detection in RSIs. In this article, we would like to propose a more general multiclass detector.

B. Context Modeling in Object Detection

Context refers to information related to the circumstances wherein a process occurs. Context benefits image understanding and interpretation and many empirical studies [38], [39] on generic object detection in ground-based scene images have shown that object detectors can be improved by proper modeling of context because they not only consider object visual appearances but also utilize *object-to-object* and *object-to-scene* relationships. Based on this, there are two types of context modeling approaches, i.e., *instance-level* context modeling and *scene-level* context modeling. In contrast to the works in [13] and [15]–[17] that detect objects individually without directly exploiting their relations, to model instance-level context, most studies focus on using object *relative position* or *co-occurrence pattern* (e.g., a mouse is near a computer). For example, Desai *et al.* [40] incorporate spatial relationship modeling into a contextual cueing by framing it as a structured prediction problem. Chen and Gupta [41] model object relationships by introducing a *spatial memory network* (SPN), which assembles object instances back into a pseudo image representation to perform object-to-object relationship reasoning. Zeng *et al.* [42] proposed a novel gated bi-directional network to encode local and contextual visual patterns from different support regions. As for the scene-level context modeling, recently, Bell *et al.* [43] propose to use spatial recurrent neural networks to explore image-level contextual information. Shrivastava and Gupta [44] proposed the use of semantic segmentation to encode such information. Xie *et al.* [45] propose a multilevel context VoteNet by considering the contextual information at both the scene level and instance level. In the field of object detection in ground-based scene images, it is broadly agreed that proper modeling of context is very useful for improving object detection, especially when object appearance features are insufficient [46], [47]. As for the remotely sensed images, they are typically obtained by sensors from a vertical view and, thus, generally capture the roof information of objects, leading to severe insufficiency of image diversity and variations [48]. Therefore, exploration of context modeling for the promotion of object detection is much more important in the literature of object detection in RSIs.

C. Data Augmentation Techniques

Data augmentation techniques, such as scaling, rotating, translating, flipping, and cropping, are commonly used to train large neural networks [49]–[51]. Although these techniques are helpful to reduce overfitting when training a model, their handcrafted designs limit their potential performance. To address this problem, *AutoAugment* [52], [53] was proposed by introducing 16 geometric transformations and formulating a novel augmentation policy for each batch of data by using reinforcement learning. However, AutoAugment is a very expensive algorithm that requires training 15 000 models. In contrast, a simple data augmentation technique termed *mixup* [54], [55] performs the interpolation of data pairs to achieve augmentation and has proven successful in mitigating adversarial perturbations. Essentially, data augmentation can be seen as a kind of implicit model regularization. Thus, to improve the performance of our proposed SRAF-Net and also by considering the insufficiency of scene diversity in RSIs, we introduce a scene disturbance process by using a simple data augmentation module between different scenes to further regularize the model and improve its robustness.

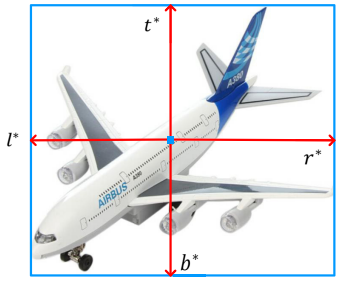


Fig. 1. Illustration of anchor-free bounding box encoding. The goal of SRAF-Net is to directly regress the 4-D vector $\mathbf{z}^* = (t^*, l^*, b^*, r^*) \in \mathbb{R}^4$, which encodes the location of a bounding box at each positive pixel.

III. Scene-Enhanced Anchor-Free Object Detection Model

To take the merit of anchor-free, our SRAF-Net follows the recent anchor-free methods and uses a wide target encoding mechanism designed for an anchor-free detection framework, which is presented in Section III-A. Considering the help of "scene related" and based on the anchor-free framework mentioned above, SE-FPN and SADH are proposed in Sections III-B and III-C for improving the detector's performance, respectively. Moreover, aim to increase the robustness and generalization of SRAF-Net; a balanced mixup data augment tech is introduced in Section III-D. The final training and inference process of SRAF-Net is described in Section III-E.

A. Anchor-Free Detection Framework

As stated above, the anchor-based methods have several drawbacks. To deal with these drawbacks, several recent anchor-free detectors use a well-designed structure to detect objects without anchors, e.g., FCOS [25] and others [26]. Following the simple network framework of them, which basically consists of a backbone, FPN, and the detection head, our proposed SRAF-Net abandons the widely used anchor mechanism for predicting the bounding box and aims at regressing a 4-D vector $\mathbf{z}_{x,y} = (l, t, r, b) \in \mathbb{R}^4$, as shown in Fig. 1, which encodes the location of a bounding box and is the distance from a location (x, y) to the corresponding

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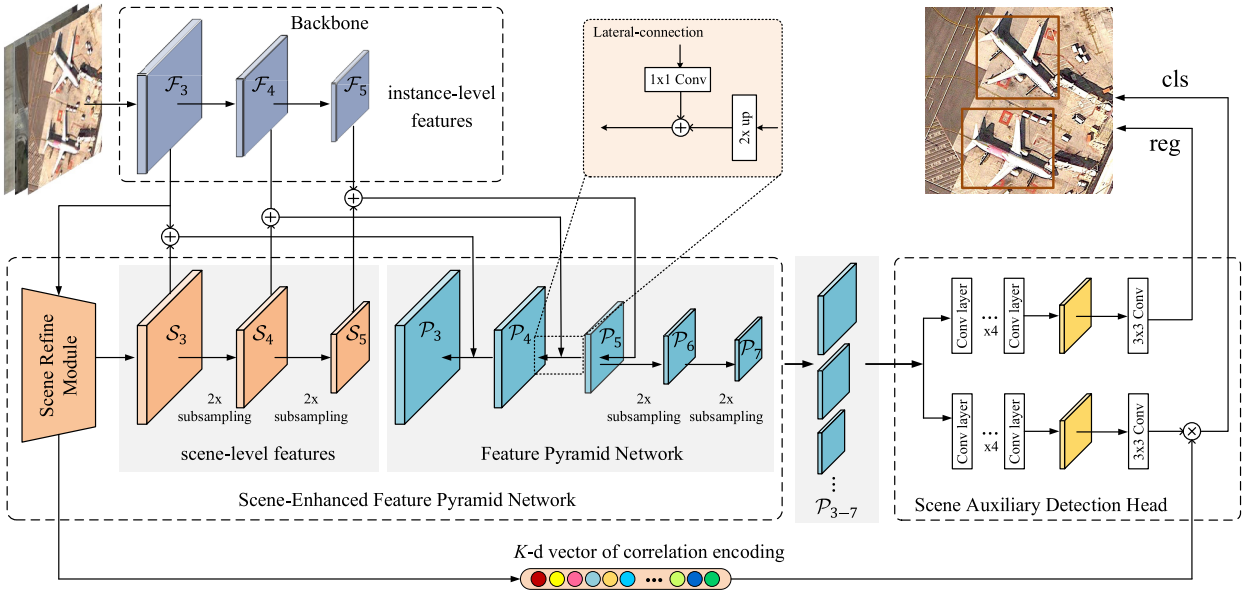


Fig. 2. Network architecture of SRAF-Net. K is the number of different categories. $\oplus$ and $\otimes$ denote a concatenation and score fusion operation at scene level and instance level defined in (3).

edges of the ground-truth box $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$, where $(x_{\min}, y_{\min})$ and $(x_{\max}, y_{\max})$ are the coordinates of the left-top and right-bottom corners of the bounding box. Then, for a positive location (x, y), the training regression targets for this location, denoted by $\mathbf{z}^*_{x,y} = (l^*, t^*, r^*, b^*) \in \mathbb{R}^4$, can be defined as

$$t^* = y - y_{\min}, \quad l^* = x - x_{\min}$$
$$b^* = y_{\max} - y, \quad r^* = x_{\max} - x. \tag{1}$$

Developed from the anchor-free detectors framework, the architecture of our proposed SRAF-Net is illustrated in Fig. 2. As the figure shows, our SRAF-Net has three parts: backbone, *SE-FPN*, and *SADH*. As for the SE-FPN, an SRM is designed to guide the network to refine multiscale contextual information (i.e., $\mathcal{S}_3$–$\mathcal{S}_5$) at a global scene level, which is then concatenated with local instance-level features (i.e., $\mathcal{F}_3$–$\mathcal{F}_5$) extracted by a backbone. Based on the concatenated multiscale and multilevel features (i.e., $\mathcal{P}_3$–$\mathcal{P}_7$) and assisted by the scene-level scores from the SRM, a *cooperative* detection head (i.e., SADH) is designed to help reduce false alarms and improve the overall performance; Sections III-B and III-C describe them in detail.

B. Scene-Enhanced Feature Pyramid Network

Several previous methods rely on the *FPN* [14] to capture multiscale local context features of objects. However, the contextual information of objects at the scene level is not fully exploited, causing low detection accuracy in some scenarios, especially for the remotely sensed images with insufficient image diversity and variations. In fact, the types of objects present are usually closely related to the scene where they are located, e.g., *ships* appear at *sea* and *vehicles* appear on *roads* with high probabilities. Inspired by this fact, and to fully integrate the global scene- and local instance-level contextual

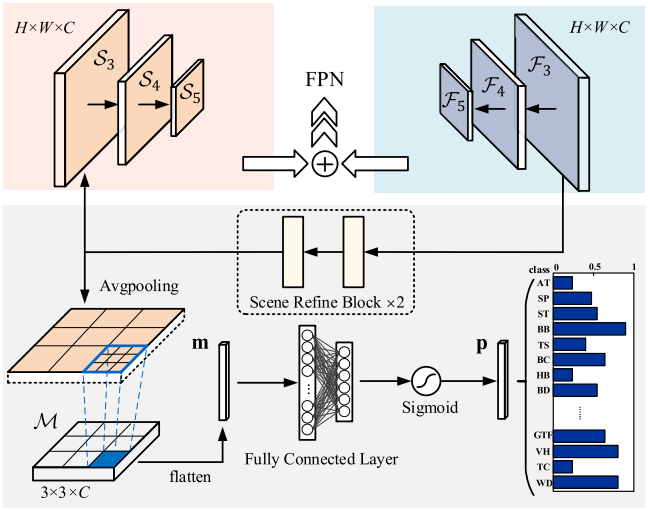


Fig. 3. Architecture of SRM.

information, we design an SRM to extract the scene-contextual feature and then develop the SE-FPN by adding our SRM into the framework of FPN.

Let $\mathcal{F}_3 \in \mathbb{R}^{H\times W\times C}$ be the extracted feature map from a backbone network (e.g., ResNet [20] on stage 3), where C, H, and W, respectively, denote the channel dimension, height, and width of feature maps, and let $K \in \mathbb{R}$ denote the number of object categories. As shown in Fig. 3, the SRM that we designed first feeds the feature map $\mathcal{F}_3$ into two *scene refine blocks* to obtain the scene-level features $\mathcal{S}_3 \in \mathbb{R}^{H\times W\times C}$, which has the same spatial size as $\mathcal{F}_3$. Then, an *average pooling* (Avgpooling) is applied on the scene-level features $\mathcal{S}_3$, resulting in a pooled feature map $\mathcal{M} \in \mathcal{R}^{3\times 3\times C}$, which is flattened into a vector $\mathbf{m} \in \mathbb{R}^{9C}$. The vector $\mathbf{m}$ is then input into a fully connected layer with a sigmoid function

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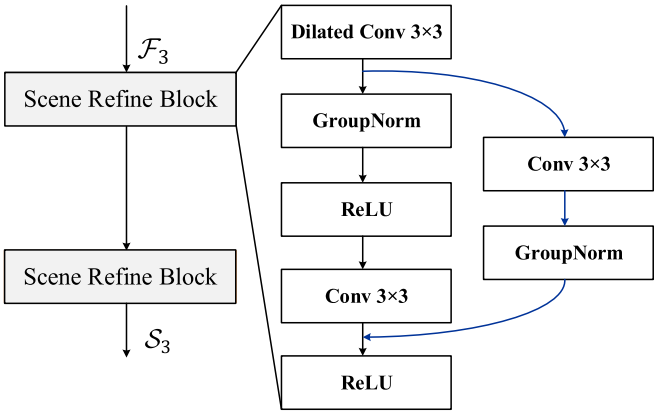


Fig. 4. Architecture of two scene refine blocks. Note that the input and output for each block have the same number of channels.

as its top activation function. Finally, the SRM outputs a probability vector $\mathbf{p} \in \mathbb{R}^K$ encoding the existence prediction scores of objects within a scene. The score vector $\mathbf{p} \in \mathbb{R}^K$ can encode the correlation between a scene and objects in a weak supervised manner by minimizing the object-to-scene correlation loss L_{osl} below.

To enforce the designed SRM focus on the most relevant semantic information at the scene level, we use two scene refine blocks that are shown in Fig. 4. As we can see, each block has a 3×3 dilated convolution [56], using a dilation rate of 3 for the first block and 1 for the second. The resulting feature maps are then passed through a GroupNorm normalization [57] and activated by a *rectified linear unit* (ReLU) nonlinear function. Finally, a normal 3×3 convolutional filter, followed by a ReLU, is employed to further extract more discriminative features of increasing abstraction at the scene level. In addition, a shortcut connection with a normal 3×3 convolution plus a GroupNorm normalization is used to alleviate the gradient vanishing problem and then benefit training.

To improve the multiscale scene-level feature extraction capabilities of FPN, we simply downsample the scene-level features $\mathcal{S}_3 \in \mathbb{R}^{H \times W \times C}$, resulting in the pyramid of two pooled scene-relevant features $\mathcal{S}_4$ and $\mathcal{S}_5$, which have the same spatial size and number of channels as the local instance-level features $\mathcal{F}_4$ and $\mathcal{F}_5$ extracted by the backbone network, e.g., ResNet [20] at stage 4 and stage 5. The scene-level features $\mathcal{S}_i$ of varying scales are concatenated with the instance-level features $\mathcal{F}_i$, i.e., $\mathcal{S}_i \oplus \mathcal{F}_i$, $i = 3, 4, 5$, and then fed into the FPN for the subsequent prediction, as shown in Fig. 2. Since this procedure integrates the scene-level features into the FPN, we name it SE-FPN.

C. Scene Auxiliary Detection Head

To perform localization and recognition of object instances, a *detection head* is usually applied on the feature maps extracted by the backbone, architecturally mirroring a head attached to a backbone. The detection head is very important for object detection tasks. However, the scene information is not directly considered and/or explored in the design of detection heads, usually leading to a high false recall rate.

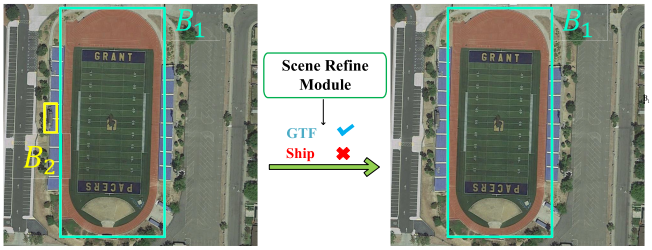


Fig. 5. Illustration of the appearance ambiguity problem, which can be alleviated by combining the scene-level information with the instance-level information.

The problem is even more acute for RSIs because only the roof information is captured, and thus, the profile information of objects is limited. A simple example is shown in Fig. 5, through traditional detection heads, and after *nonmax suppression* (NMS) operator, we obtain the final correct predicted box B_1 with $c_1 = 0.9$; however, due to the appearance of a roof similar to that of a ship, a false predicted result B_2 with $c_2 = 0.6$ for *ships* is given as well. Since its scale is similar to that of ordinary ships, it is difficult to suppress it by means of statistical scales. However, as we know, ships are almost impossible to appear next to the ground track field. In fact, this *appearance similarity* problem in RSIs can be alleviated by making use of the scene information. Specifically, if we have the scene-level confidences obtained by using the SRM, e.g., the score $p_1 = 0.8$ for ground track field and $p_2 = 0.3$ for ships in this image, we can reduce the scores of box B_2 by a combination of the instance-level score and the scene-level score, e.g., $\tilde{c}_2 = 0.6c_2 + 0.4p_2 = 0.48$. Motivated by using scene information to suppress false detection, we propose an *SADH*, which uses the outputs of the SRM as a scene auxiliary for further refinement of object classification.

Similar to the detection heads in [25] and [34], the proposed SADH is also composed of two subnetworks: one for regression of the location and another for classification of probability prediction of objects' presence at each spatial position for each of the K object categories. As shown in Fig. 6, each subnet is a small *fully convolutional network* (FCN) attached to each pyramid level $\mathcal{P}_i, i = 3, \ldots, 7$, and uses four 3×3 convolutional layers, each followed by ReLU activations. Finally, the sigmoid and exp(·) functions are attached to the top of the classification subnet and regression subnet, respectively.

We also incorporate the vector $\mathbf{p} = (p_1, p_2, \ldots, p_K)^T$, which is obtained by the SRM and serves as an encoding of the correlation between a scene and object categories, into the classification subnet. As shown in Fig. 6, to be computationally efficient, the scene auxiliary vector $\mathbf{p}$ is thresholded and gets the scene-level indicator $\mathbf{h} = (h_1, h_2, \ldots, h_K)^T$ by

$$h_i = \begin{cases} 1, & p_i > \tau \\ 0, & p_i \le \tau \end{cases} \quad i = 1, 2, \ldots, K \tag{2}$$

where τ is a threshold parameter. With the help of the scene-level indicator $\mathbf{h}$, the scores $\mathcal{C} \in \mathbb{R}^{H \times W \times K}$ of the classification subnet can be adjusted by the following score

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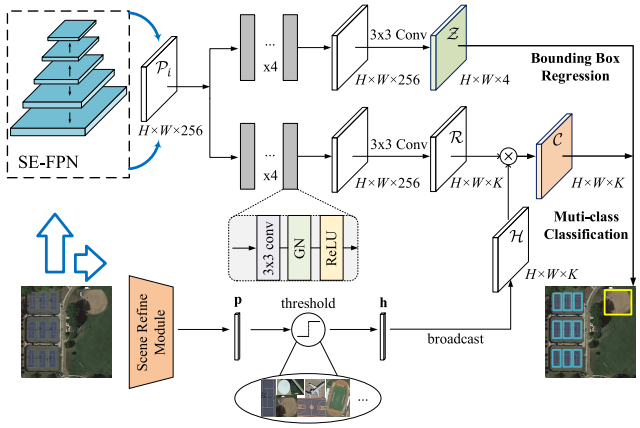


Fig. 6. Architecture of our designed *SADH*. $\otimes$ denotes the scene auxiliary process in (3).

fusion operator between instance level and scene level as

$$\mathcal{C} = \alpha\mathcal{R} + (1-\alpha)\mathcal{H} \tag{3}$$

where $\mathcal{H} \in \mathbb{R}^{H\times W\times K}$ is the broadcast version of $\mathbf{h} \in \mathbb{R}^{K}$, $\mathcal{R} \in \mathbb{R}^{H\times W\times K}$ is the scores output by the traditional classification subnet, and α is a tradeoff parameter to balance the scene- and instance-level scores. The proposed SADH can assist the network in discriminating ambiguous samples with similar appearances but different categories.

D. Balanced Mixup Data Augment

Data augmentation is a simple yet effective technique to increase the amount of data and its diversity. Considering scene transplantation, diversity, and complexity, we propose a *balanced mixup data augment* (BMDA) to increase the robustness and generalization of SRAF-Net. Specifically, the BMDA can be seen as a special case of the mixup data augmentation technique [54], [55], which creates synthetic samples from existing data by mixing up pixels of two image samples and their labels. We found that a complex scene can often be composed of two simple scenes, e.g., an airplane is with vehicles, as shown in Fig. 7. For this reason, unlike the mixup [54], which randomly sets the blending weight drawn from a beta distribution, the BDMA simply takes a balanced weight to produce virtual images with more complicated scenarios, i.e., constructs virtual training example $\widetilde{\mathcal{X}}$ based on two randomly selected samples $\mathcal{X}_i$ and $\mathcal{X}_j$ using the following formula:

$$\widetilde{\mathcal{X}} = \mu\mathcal{X}_i + (1-\mu)\mathcal{X}_j \tag{4}$$

where $\mu = 0.5$, resulting in a balanced interpolation between $\mathcal{X}_i$ and $\mathcal{X}_j$. As shown in Fig. 7, the image pixels are mixed up with a balanced weight and the object labels are merged as new one-hot class labels $\widetilde{\mathcal{C}}^*$. In this way, the diversity of scenes may be improved, leading to increased robustness of the transplanted objects and improved generalization. The BMDA can boost the performance of our SRAF-Net without introducing extra computational costs.

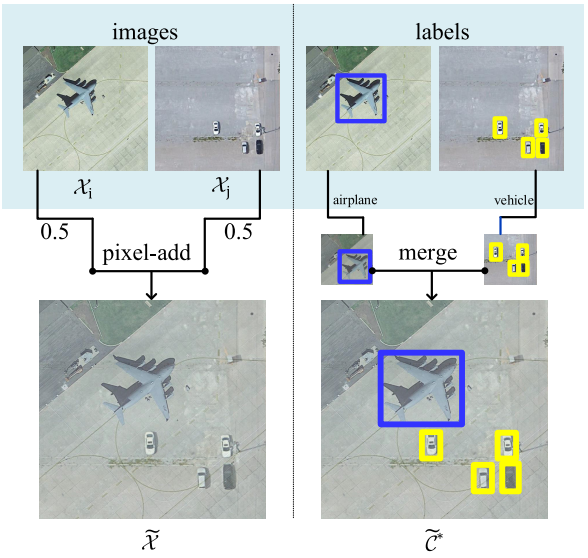


Fig. 7. Illustration of BMDA: for randomly selected two different images, BMDA creates another mixup result for both images and labels.

E. Training and Inference

1) Training: The anchor-based method selects a training sample for positive and negative with the help of candidate anchors and the IoU distance between them and the ground-truth box. Instead, in the training process of SRAF-Net, followed by FCOS [25], we treat each pixel at each pyramid level $\mathcal{P}_i, i = 3, \ldots, 7$, as training samples. Specifically, a location (x, y) on the feature maps of $\mathcal{P}_i$ is regarded as a positive sample if it falls into a ground-truth box, and its class label $\mathbf{c}^*_{x,y} \in \{0,1\}^K$ is assigned to be the class label of the ground-truth box. Otherwise, it is considered to be a negative sample with its class label $\mathbf{c}^*_{x,y} = \mathbf{0} \in \{0,1\}^K$.

2) Loss Function: The SRAF-Net can be trained by minimizing three loss terms, i.e., object category classification loss L_{cls}, object bounding box regression loss L_{reg}, and object-to-scene correlation loss L_{osl}. For each training image sample, the overall loss function is formulated as

$$\begin{aligned} L\big(\{\mathbf{c}_{x,y}\},\{\mathbf{z}_{x,y}\},\{\mathbf{p}\}\big) = &\frac{\lambda_1}{N_{\text{pos}}}\sum_{x,y} L_{\text{cls}}\big(\mathbf{c}_{x,y},\mathbf{c}^*_{x,y}\big) \\ &+\frac{\lambda_2}{N_{\text{pos}}}\sum_{x,y}\mathbb{1}_{\{\mathbf{z}^*_{x,y}>0\}} L_{\text{reg}}\big(\mathbf{z}_{x,y},\mathbf{z}^*_{x,y}\big) \\ &+\lambda_3 L_{\text{osl}}\big(\mathbf{p},\mathbf{p}^*\big) \end{aligned} \tag{5}$$

where $L_{\text{cls}}(\mathbf{c}_{x,y},\mathbf{c}^*_{x,y})$ is the well-known focal loss as in [34] at location (x, y), $L_{\text{reg}}(\mathbf{z}_{x,y},\mathbf{z}^*_{x,y})$ is the *generalized IoU* (GIoU) loss as in [58] at location (x, y), and $L_{\text{osl}}(\mathbf{p},\mathbf{p}^*)$ is a *binary cross-entropy* (BCE) loss function to determine whether the object category appears in the scene. $\mathbf{p}^*$ is the image-level label that notifies which categories appear in the image or which is a zero/one paradigm. N_{pos} is the total number of positive samples, and $\mathbb{1}_{\{\mathbf{z}^*_{x,y}>0\}}$ is an indicator function (0 for negative samples and 1 for positive samples). We simply set the hyperparameters $\lambda_1 = \lambda_2 = \lambda_3 = 1$ in this study. Note that the summation is calculated over all locations (x, y) on five levels of feature maps, i.e., $\mathcal{P}_{3-7}$.

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3) Inference: The inference is performed by forwarding an input image through the network. For the SE-FPN branch, we obtain the feature maps at different pyramid levels $\mathcal{P}_{3-7}$ and scene-level auxiliary scores $\mathbf{p}$. Based on these, the SADH products the classification scores $\mathcal{C} = (\mathbf{c}_{x,y}) \in \mathbb{R}^{H\times W\times K}$ and the location regression predictions $\mathcal{Z} = (\mathbf{z}_{x,y}) \in \mathbb{R}^{H\times W\times 4}$ at each pyramid level. We choose the locations (x, y) by thresholding the classification scores $\mathbf{c}_{x,y}$ by 0.05 as positive samples and then inverting their 4-D distance vector $\mathbf{z}_{x,y}$ based on (1), followed by *nonmaximum suppression* (NMS) [59], to obtain the final bounding boxes.

IV. Experiments

To validate the effectiveness of our proposed SRAF-Net detection framework, we evaluated it on three publicly available benchmark datasets: the NWPU VHR-10 dataset [60], the DOTA dataset [61], and the DIOR dataset [48]. The proposed SRAF-Net is also compared with other SOTA methods, e.g., [13]–[16], [22], [23], [34], and [62]. All the methods were implemented on a Linux operating system (Ubuntu 18.04.1) with a Core i7-9700 CPU, 32-GB RAM, and two NVIDIA GTX-1080Ti GPUs (11-GB memory).

A. Dataset

1) NWPU VHR-10 Dataset [60]: This is the most widely used dataset for object detection in RSIs. It contains a total of ten categories of objects, 800 *very-high-resolution* (VHR) optical RSIs, including 715 RGB color images from *Google Earth* with a resolution of 0.5–2.0 m. The other 85 images are pansharpened infrared images with a spatial resolution of 0.8 m. These ten categories are *airplane* (AL), *baseball diamond* (BB), *basketball court* (BC), *bridge* (BD), *ground track field* (GTF), *harbor* (HB), *ship* (SP), *storage tank* (ST), *tennis court* (TC), and *vehicle* (VH). Of all the samples, 150 negative samples that did not contain any objects were not considered for the experiment. Therefore, only 650 positive samples were used. We first randomly divided the dataset into a training set, a validation set, and a test set according to the ratio of 4:3:3, which contained 260, 195, and 195 images, respectively. For each divided set, the appropriate proportion of categories was assigned.

2) DOTA Dataset [61]: DOTA is a large-scale dataset for object detection in aerial images. It contains 2806 aerial images from different sensors and systems. The size of each image ranges approximately 800×800 to 4000×4000 pixels. This dataset contains 15 common object categories and 188 282 instances that are annotated by an arbitrary quadrilateral. These 15 categories are *plane* (PL), *ship* (SP), *storage tank* (ST), *baseball diamond* (BD), *tennis court* (TC), *basketball court* (BC), *ground track field* (GTF), *harbor* (HA), *bridge* (BR), *large vehicle* (LV), *small vehicle* (SV), *helicopter* (HC), *roundabout* (RA), *soccer ball field* (SBF), and *basketball court* (BC). Half of these images are randomly selected as training set, 1/6 as validation set, and 1/3 as test set. Here, we only pay particular attention on the *horizontal bounding box* (HBB). We divide the images into 800×800 subimages with an overlap of 200 pixels and select the external rectangle of the bounding boxes as the ground truth.

3) DIOR Dataset [48]: This dataset was recently published for the object detection in RSIs. It is quite challenging and contains 23 463 optical RSIs and 190 288 objects, with their object bounding boxes manually marked. The train and validation sets contain 5862 and 5863 images, respectively; then, the remaining 11 738 images are used as the test set. The spatial resolution for this dataset ranges from 0.5 to 30 m. It has a total of 20 common ground object categories, namely, *airplane* (c1), *airport* (c2), *baseball field* (c3), *basketball court* (c4), *bridge* (c5), *chimney* (c6), *dam* (c7), *expressway service area* (c8), *expressway toll station* (c9), *harbor* (c10), *golf course* (c11), *ground track field* (c12), *overpass* (c13), *ship* (c14), *stadium* (c15), *storage tank* (c16), *tennis court* (c17), *train station* (c18), *vehicle* (c19), and *windmill* (c20). Each image is of size 800×800 pixels and with excellent scale variability, e.g., the scale of a vehicle is only below 100, but an overpass can reach 800 at most. The environment and weather at the time of image capture also vary greatly, making it one of the most challenging datasets for evaluating the performance of object detection models.

B. Evaluation Metrics

To comprehensively assess the performance of object detection approaches, we adopt the widely used *average precision* (AP) quantitative evaluation metric. We also calculate the AP and *mean AP* (mAP) under the same threshold. AP is achieved by calculating the area under the *precision–recall* (P–R) curve. The precision and recall are defined as follows:

$$\text{Precision} \triangleq \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{6}$$

$$\text{Recall} \triangleq \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{7}$$

where TP, FP, and FN denote the *true positives*, *false positives*, and *false negatives*, respectively. If the IoU value between a predicted bounding box and its ground-truth is greater than 0.5, the box is regarded as a TP; otherwise, it is assigned to be a FP. A predicted box only matches one ground-truth box, and the rest are considered as FP. The mAP is calculated by the mean value of AP over all categories.

C. Comparison and Implementation Details

To better evaluate the performance of our comparison method, we reproduce some popular SOTA methods on those datasets, including the two-stage methods, such as Faster R-CNN [13], FPN [14], and Cascade R-CNN [15], the one-stage anchor-based method, such as SSD [16], YOLOv3 [62], and RetinaNet [34], and several modern anchor-free detectors, such as FCOS [25], the key-point representative methods, CornerNet [22], and CenterNet [23]. The evaluation results are reported in Tables I–III; without special emphasis, the symbol * indicates that the method is trained with simple data augmentation techniques (e.g., flip and expand), and the symbol † indicates that the method is trained with complex data augmentation techniques (e.g., extend random mirror, random crop, scale, color, and photometric distort).

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TABLE I
PERFORMANCE COMPARISON ON THE NWPU VHR-10 DATASET

Method	Model	Backbone	AL	SP	ST	BB	TS	BC	GTF	HB	BD	VH	mAP
Anchor-Based	Faster R-CNN*	VGG16	90.82	80.34	49.54	89.30	69.96	81.82	98.97	68.18	27.92	58.58	71.54
		ResNet-101	90.53	79.81	51.21	89.09	70.72	81.82	90.91	70.79	42.88	56.56	72.43
	Faster R-CNN with FPN*	ResNet-50	90.75	**85.38**	82.35	88.43	73.60	51.98	89.09	62.90	61.95	65.73	68.38
		ResNet-101	90.71	84.40	82.79	90.22	80.67	74.67	98.98	80.32	68.79	80.94	75.68
	Cascade R-CNN with FPN*	ResNet-50	90.41	76.49	75.37	90.29	61.92	22.99	95.27	72.54	56.56	55.80	63.42
		ResNet-101	90.91	78.21	83.39	90.47	78.95	68.11	98.49	**81.82**	**72.73**	81.33	74.95
	YOLOv3†	Darknet-53	92.50	62.90	56.12	95.30	64.08	47.99	92.17	70.06	59.48	72.59	71.32
	SSD†	VGG16	90.44	73.43	**87.14**	90.02	46.29	39.98	99.51	80.22	26.31	69.73	70.31
	RetinaNet†	ResNet-50	**98.73**	81.90	74.47	95.83	79.26	90.90	97.68	77.24	67.17	69.04	83.22
Anchor-Free	FCOS*	ResNet-50	92.74	77.17	75.17	**98.47**	70.48	71.19	89.34	39.31	49.89	86.55	75.00
	CornerNet†	Hourglass-104	72.10	53.67	56.51	92.67	67.79	83.16	**99.96**	49.17	44.13	77.25	69.64
	CenterNet†	Hourglass-104	73.09	71.97	59.91	87.41	65.12	73.37	95.27	75.48	53.76	55.75	71.11
	SRAF-Net*	ResNet-50	*94.59*	83.80	72.84	*96.95*	**88.39**	**92.28**	98.95	63.53	53.99	**89.21**	**83.45**

TABLE II
PERFORMANCE EVALUATION OF HBB TASK ON THE DOTA DATASETS

	Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Anchor-based	RICNN	62.10	26.00	20.10	28.50	30.40	28.40	41.30	51.70	43.10	27.10	25.30	28.10	31.50	26.50	13.40	32.20
	R-FCN	81.01	58.96	31.64	58.97	49.77	45.04	49.29	68.99	52.07	67.42	41.83	51.44	45.15	53.30	33.89	52.58
	FR-H	80.32	77.55	32.86	68.13	53.66	52.49	50.04	90.41	75.05	59.59	57.00	49.81	61.69	56.46	41.85	60.46
	FPN	88.70	75.10	52.60	59.20	69.40	78.80	**84.50**	90.60	81.30	82.60	52.50	62.10	76.60	66.30	60.10	72.00
	ICN	90.00	77.70	53.40	**73.30**	73.50	65.00	78.20	90.80	79.10	84.80	57.20	62.10	73.50	70.20	58.10	72.50
	SCRDet	**90.18**	**81.88**	**55.30**	73.29	72.09	**77.65**	78.06	**90.91**	**82.44**	**86.39**	64.53	63.45	75.77	**78.21**	60.11	**75.35**
	SSD	57.85	32.79	16.14	18.67	0.05	36.93	24.74	81.16	25.10	47.47	11.22	31.53	14.12	9.09	0.00	29.86
	YOLOv2	76.90	33.87	22.73	34.88	38.73	32.02	52.37	61.65	48.54	33.91	29.27	36.83	36.44	38.26	11.61	39.20
	YOLOv3	77.73	62.11	37.32	45.16	64.34	49.95	77.58	89.90	68.12	71.64	29.79	50.98	62.37	50.71	44.22	58.79
	AFANet	89.42	73.89	47.33	59.86	64.49	67.33	82.92	90.69	66.27	72.33	67.62	62.17	**76.82**	60.49	52.81	68.97
	GLNet	89.40	71.50	43.10	68.40	31.90	52.50	53.90	79.90	66.90	77.20	**74.10**	**64.60**	73.10	59.20	**74.00**	65.30
	ASBL-RetinaNet	89.51	74.07	46.91	55.54	**73.78**	66.87	78.48	90.86	70.09	73.20	46.71	61.34	70.50	72.17	32.84	66.86
Anchor-free	CornerNet†	67.85	**78.94**	53.59	**68.05**	31.39	46.52	27.08	87.96	63.75	53.57	62.46	69.79	58.79	43.06	23.94	55.78
	CenterNet†	88.43	71.89	54.64	67.36	40.92	61.11	43.96	90.51	59.44	63.09	**63.23**	70.33	68.87	52.62	28.35	61.65
	FCOS†	88.81	71.63	**56.35**	40.86	49.37	74.58	68.92	89.56	67.31	70.77	44.63	**70.97**	66.90	42.71	36.97	62.69
	DPAFPN	80.23	73.17	48.81	55.13	55.97	49.06	57.76	81.12	66.57	75.75	45.45	65.82	64.88	**72.38**	**51.36**	62.92
	CenterFPANet	88.83	69.63	47.22	45.80	49.96	73.89	63.46	90.44	59.52	**83.65**	58.40	64.25	69.43	55.00	40.55	64.00
	SRAF-Net†	**88.93**	72.76	50.10	45.93	**59.51**	**75.69**	**83.77**	**93.00**	**70.32**	67.08	55.63	62.69	**71.45**	47.36	41.80	**65.73**

1) Implementation Details of the Comparisons: In the experiments, the *ROI align* operation [63] was used to improve the region proposal process. In the two-stage approaches, the initial learning rate is set to 0.001 with *stochastic gradient descent* (SGD). The other hyperparameters (e.g., the scale and aspect ratio of anchors) strictly follow the corresponding works [13]–[15], [63]. A multiscale training is adopted in the implementation of YOLOv3 [62]. When evaluating RetinaNet [34], we set the learning rate to 0.001 for stable training, rather than the original 0.01, and set parameters $\alpha = 0.25$ and $\gamma = 2$ in the focal loss. The optimizer for the one-stage models is SGD as well, and its momentum is set to 0.9 with 0.0005 weight decay. As usual, we use the pretrained weights in ImageNet [64] as the backbone initialized weights. The CornetNet [22] and CenterNet [23] are set with a batch size of 8 and trained probably 30k, 160k, and 150k iterations for the three datasets, respectively.

2) Implementation Details of SRAF-Net: There are several parameters in our proposed SRAF-Net, including the L_{cls} part in (5), where we set its hyperparameters to be the same as the focal loss in [34]. All the *normalization* layers used in our SRAF-Net are the GroupNorm [57]. Specifically, when training our models, the tradeoff parameter α in (3) is set to 0.95. The initial learning rate is set to 0.0001 and multiplied by 0.1 after half of the total iterations. Empirically, we use Adam as the optimizer. Except for the backbone networks, the additional layers are initialized by following ResNet [20]. The training batch size for SRAF-Net is set to 4, and we train the model with 13k iterations for the NWPU VHR-10 dataset, 150k for the DOTA, and 130k for the DIOR dataset. For the inference stage, τ is set to 0.001 in (2) in our SRAF-Net. The thresholds used in NMS and for the final output scores were 0.5 and 0.05, respectively.

D. Evaluation on the NWPU VHR-10 Dataset

Table I reports the AP and mAP scores on the NWPU VHR-10 dataset. As we can see from this table, our proposed SRAF-Net achieves the best detection accuracy of 83.45% mAP due to the utilization of scene-level contextual information. The RetinaNet [34] obtains a comparable result of 83.22% mAP, benefiting from a dense sampling of possible object locations and the effectiveness of focal loss in dealing with the class imbalance. Both SRAF-Net and RetinaNet outperform the SOTA two-stage methods, such as Faster R-CNN [13], Faster R-CNN with FPN [14], and Cascade R-CNN with FPN [15], and are also better than the SOTA one-stage anchor-based methods, such as SSD [16] and

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TABLE III
COMPARISON WITH NINE SOTA METHODS IN TERMS OF AP (%) AND MAP (%) ON THE DIOR DATASET

Method	Model	Backbone	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20	mAP
Anchor-Based	Faster R-CNN*	VGG16	77.7	62.1	81.4	80.9	35.8	81.5	43.8	69.0	57.8	67.7	78.8	33.8	56.8	22.2	88.8	38.3	80.7	41.8	31.4	80.5	60.5
		ResNet-101	75.3	75.2	81.2	80.3	37.9	81.4	60.0	79.2	57.6	75.8	77.2	36.0	56.1	23.0	88.1	38.9	80.7	51.0	32.6	78.2	63.3
	Faster R-CNN with FPN*	ResNet-50	80.8	65.9	81.0	78.4	36.2	72.5	53.2	71.1	61.5	65.4	75.1	39.8	55.9	68.7	88.2	52.2	81.5	39.1	42.8	78.4	61.3
		ResNet-101	81.1	76.9	89.8	80.5	45.1	81.7	63.4	80.9	63.2	76.3	78.3	43.3	58.4	68.5	89.1	59.0	81.1	**60.2**	47.8	79.3	66.9
	Cascade R-CNN with FPN*	ResNet-50	81.3	75.2	81.2	78.6	39.2	73.8	55.1	77.0	62.1	66.8	77.3	39.5	57.7	70.1	87.8	52.9	81.7	38.2	45.0	80.5	62.9
		ResNet-101	81.2	**81.4**	90.1	81.1	**46.9**	81.5	**68.1**	84.2	65.3	74.5	81.6	37.8	**62.5**	68.9	88.8	60.5	81.0	57.5	47.7	80.7	67.7
	YOLOv3†	Darknet-53	77.0	50.6	82.4	70.9	19.3	79.4	35.9	61.9	44.7	59.4	47.6	45.4	44.9	**72.8**	72.4	**66.7**	77.1	53.9	33.0	75.6	58.5
	SSD†	VGG16	63.5	74.4	78.5	77.3	30.7	76.5	61.8	71.0	57.0	72.9	69.2	**52.8**	54.3	32.8	85.9	32.3	80.7	43.9	25.4	57.1	59.9
	RetinaNet (600)†	ResNet-50	80.0	54.6	81.1	85.0	21.1	77.8	35.7	59.9	52.2	70.0	71.9	33.7	43.7	46.4	85.3	41.0	89.4	16.2	24.7	67.5	56.9
		ResNet-101	80.9	57.8	82.2	87.0	24.3	78.7	42.2	71.9	55.4	75.9	75.4	39.2	46.3	46.8	87.5	40.2	90.1	20.6	25.3	70.6	59.9
	RetinaNet (800)†	ResNet-50	87.8	59.0	87.7	87.6	25.6	81.8	35.8	72.1	57.4	75.5	79.0	37.5	48.5	70.2	87.7	55.2	93.2	25.2	36.1	81.1	64.2
		ResNet-101	87.9	66.7	89.6	87.8	29.6	86.7	46.7	75.4	59.1	75.8	82.4	36.7	48.4	69.2	89.7	57.8	93.9	25.3	37.9	84.7	66.6
Anchor-Free	FCOS*	ResNet-50	73.6	73.2	84.3	77.4	32.1	76.6	52.4	62.0	58.4	63.1	71.9	17.1	46.1	51.1	71.4	46.7	82.4	37.2	39.7	82.7	60.0
	CornerNet†	Hourglass-104	68.5	77.1	85.2	78.7	46.9	76.9	60.2	73.9	**74.8**	68.6	79.1	16.8	58.9	34.5	89.1	40.0	84.7	52.3	45.0	70.1	64.1
	CenterNet†	Hourglass-104	84.6	73.8	90.7	82.2	46.7	80.3	54.6	72.6	74.1	68.4	**83.7**	27.1	60.7	51.5	**91.4**	54.7	87.2	50.6	55.4	85.0	68.8
	SRAF-Net*	ResNet-50	85.8	*80.6*	*91.6*	87.2	39.0	81.3	52.6	81.7	68.4	71.9	82.0	14.2	55.2	59.5	80.5	55.4	**93.3**	50.1	**56.9**	**92.9**	*69.0*
		ResNet-101	**88.4**	76.5	**92.6**	**87.9**	35.8	**83.8**	58.6	**86.8**	66.8	**76.4**	82.8	16.2	58.0	59.4	80.9	55.6	90.6	52.0	53.2	*91.0*	**69.7**

YOLOv3 [62], by a large margin of almost 10% mAP. Specifically, in terms of *tennis court*, *basketball court*, and *vehicles* categories, our SRAF-Net achieves the best detection accuracy among all the methods. The SRAF-Net performs the second best for the *airplane*, *baseball diamond*, and *ground track field* categories. In addition, SRAF-Net has the advantages of low training complexity and fast inference speed compared with the two-stage detectors. As also can be seen from Table I, the proposed SRAF-Net shows much better performances than the anchor-free counterparts, i.e., CornerNet [22], and CenterNet [23], with significant improvement gains.

E. Evaluation on the DOTA Dataset

On the DOTA dataset, Table II shows the performance of our method on the HBB prediction task with numerous recent methods, including RICNN [18], FR-H [13], R-FCN [65], FPN [14], ICN [66], SCRDet [67], SSD [16], YOLOv2 [68], YOLOv3 [62], AFANet [69], GLNet [70], ASBL-RetinaNet [71], CornerNet [22], CenterNet [23], FCOS [25], DPAFPN [27], and CenterFPANet [28]. Based on Table II, we can find that our SRAF-Net obtains an mAP of 65.73%, which outperforms some popular anchor-based detectors, e.g., RICNN, R-FCN, FR-H, SSD, YOLOv3, and GLNet. Among anchor-based methods, the robust detector SCRDet gets a higher mAP of 75.35%; however, it is designed to solve the problem of low recall of small objects and anchor sampling caused by insufficient anchor scale. This further showed that it is very difficult for the anchor-based methods in the setting of handcrafted designs, such as anchor scale and ratio. As for the other anchor-free methods, the recent keypoint-based methods, CornerNet and CenterNet, can get 55.78% mAP and 61.65% mAP, respectively. The CenterFPANet, which is designed based on CenterNet with attention mechanism, can achieve a relatively higher mAp of 64.00%. On the other hand, FCN-based detector FCOS and DPAFPN give a similar mAP of 62.69% and 62.92%. Above all, by making use of the scene-relevant information, our SRAF-Net shows relatively better detection performances than the SOTA anchor-free methods and obtains the best AP for more than half the number of categories.

F. Evaluation on the DIOR Dataset

As shown in Table III, we present a comparison between our SRAF-Net and the SOTA detectors on the DIOR dataset; note that some reproduced results are close to the benchmark [48]. As we can see from this table, our SRAF-Net with ResNet-50 or ResNet-101 as the backbone is better than anchor-based methods, such as Faster R-CNN [13], Faster R-CNN with FPN [14], Cascade R-CNN with FPN [15], YOLOv3 [62], RetinaNet [34], and the other anchor-free methods, such as FCOS, CornerNet [22], and CenterNet [23]. From this table, we can also find that our SRAF-Net shows a relatively good detection performance for small- and large-scale objects, such as small objects, *vehicle* of 56.9% mAP and *windmill* of 92.9% mAP, and large objects, *baseball field* of 92.6% mAP, *basketball court* of 87.9% mAP, *golf course* of 76.4% mAP, *tennis court* of 93.3% mAP, and *expressway-service-area* of 86.8% mAP. In this case, the representative one-stage detectors, e.g., SSD, YOLOv3, and RetinaNet with a resolution of 600, i.e., RetinaNet (600), show relatively poor performance, while the two-stage detectors seem to perform well on a complex multiclass dataset, especially for the Cascade R-CNN with FPN, which achieves a comparable accuracy of 67.7% mAP to that of the SEAF-Net. However, they suffer from the design difficulties of anchors and low inference speed. As for the anchor-free detectors, except for our SRAF-Net, the CenterNet outperforms the others in a majority of categories, while its corner matching process is resource-intensive and slow to converge, causing a problem of high computational complexity.

Through the above verification, our SRAF-Net considers the scene contextual information and focuses on exploring the implicit relevance between objects, objects, and the scene, resulting in accurate object detection. A brief analysis of "scene-relevance" is found in the following ablation study. It must be pointed out that it is different from some of the latest

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Fig. 8. Visualization of detection results of the SRAF-Net on the DIOR dataset. As shown in the figure, SRAF-Net works well not only with extremely small objects (e.g., vehicles) but also very large objects (e.g., various kinds of courts).

remote sensing object detection studies that focus on designing complex techniques to solve the rotation invariance of certain categories of objects, e.g., *Fourier-based rotation-invariant feature boosting* [72] and ORSIm detector [73], which integrates diverse channel features' extraction. Our SRAF-Net is more simple and direct.

Compared with these SOTA detectors, SRAF-Net is anchor-free, fast, and more convenient for inference, and its intuitive detection results on the DIOR dataset are shown in Fig. 8. As shown in the visualization examples, our SRAF-Net performs well on both small and large objects.

G. Ablation Study

In this section, we present some ablation studies conducted to evaluate the contribution of several important elements to our SRAF-Net, i.e., SE-FPN, SADH, and the BMDA, in terms of NWPU VHR-10, DOTA, and DIOR datasets. For all ablation studies, unless otherwise specified, we use an FCN-based anchor-free model that only covers a generic detection head, similar to FCOS without centerness, as the ***Baseline***.

To investigate the effect of scene-contextual information explored by our SRAF-Net, we construct its variants, denoted by SRAF-Netv1 and SRAF-Netv2 as the SRAF-Net removing the SADH and/or BMDA parts, respectively. The results have been reported in Table IV. For the NWPU VHR-10 dataset, the SE-FPN, SADH, and BMDA contributed to the ***Baseline*** on a performance improvement of 3.51%, 2.97%, and 3.57% mAP, respectively. For the DIOR dataset, the above methods can give an improvement of mAP at 5.83%, 1.4%, and 3.38%,

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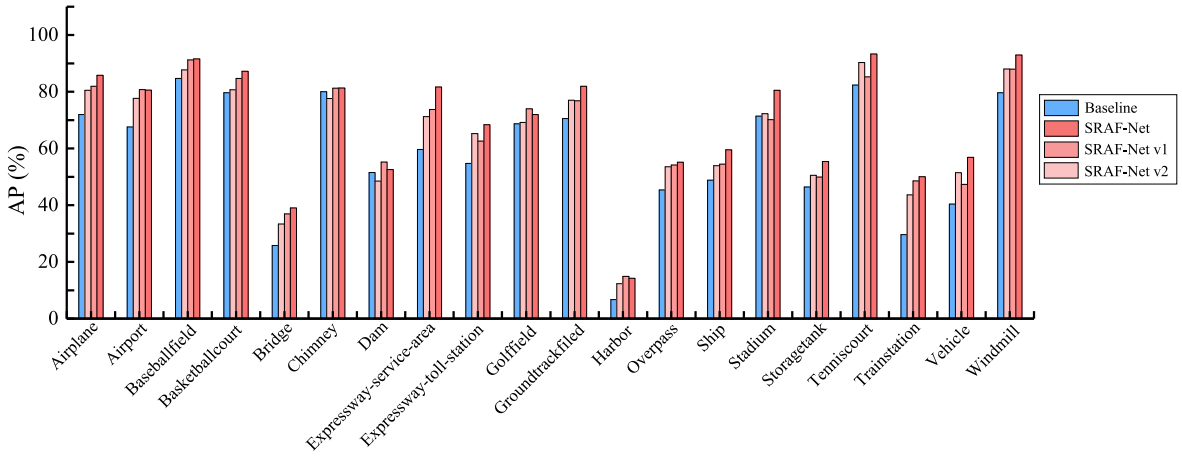


Fig. 9. AP for all categories on the DIOR dataset. We report performances for the baseline, SRAF-Net, and its two variants (SRAF-Netv1 and SRAF-Netv2, which are obtained by removing the SADH and/or BMDA parts).

TABLE IV

EFFECTIVENESS OF THREE SCENE INFORMATION RELEVANT PARTS, i.e., SE-FPN, SADH, AND BMDA, IN OUR SRAF-NET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

	SRAF-Net	SRAF-Netv1	SRAF-Netv2	***Baseline***
SE-FPN	✓	✓	✓	
SADH	✓	✓		
BMDA	✓			
NWPU VHR-10	**83.45**	79.88	76.91	73.40
DOTA	**65.73**	61.64	60.89	60.21
DIOR	**69.01**	65.63	64.23	58.30

which has demonstrated their effectiveness. A more detailed performance improvement is illustrated in Fig. 9. On the more challenging DOTA dataset, we notice that our proposed methods have obtained a positive gain, which promotes the ***Baseline*** of mAP of 0.68%, 0.75%, and 4.09%, respectively. This fully demonstrates the significant importance of encoding the scene contextual information into the detectors.

1) Analysis of Scene-Relevant Effect: In order to more intuitively role the scene-relevant feature extracted by our proposed SRM module, we visualized different scenario features, as shown in Fig. 10. As shown in the figure, with the different scenes (e.g., seaside and airport), our SRM module can extract relevant features of different categories, which helps the SE-FPN module to enhance feature representations and improve detection accuracy. On the other hand, our SADH can also code the correlation between categories in different scenes within the scene level. For example, in Fig. 10, when the harbor appears in the scene, the probability of ships that existed in the scene is usually high and the same for the bridge and vehicles. This correlation is first encoded by our SADH and then further assists the detection.

In order to find the importance of "scene-relevant" in the detection process, we evaluated the improvement of detectors performance by the selection of different α in (3). Employing ResNet-101 as the backbone, the evaluation result on the DIOR dataset of different α's is reported in Table V. As its shows, the "scene-relevant" is an auxiliary role, which can be proved by the performance of detectors declines as the weight α becomes higher. Empirically and depending on the ablation study, we set α equal to 0.95 in our proposed SRAF-Net.

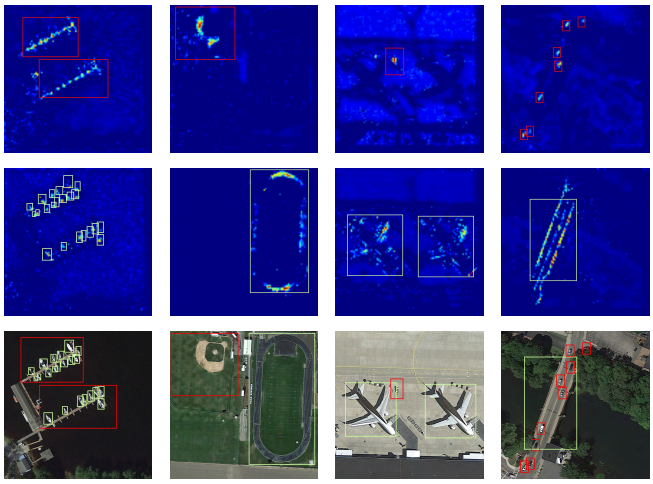


Fig. 10. Visualization of the feature maps extracted by the SRM. The scene from (Left) to (Right) contains (a) harbor and ships, (b) baseball field and ground track field, (c) airplane and vehicle, and (d) bridge and vehicle.

TABLE V

EVALUATION OF DIFFERENT α'S IN (3)

Backbone	α	**Mean AP**	**Iterations**
ResNet-101	**0.95**	**69.7**	∼130K
	0.90	65.9	∼130K
	0.85	65.2	∼130K
	0.80	64.6	∼130K
	0.75	63.9	∼130K

2) Analysis of BMDA: We conducted ablation experiments for the comparison of BMDA and common data augmentation (CDA) techniques on the DOTA dataset. The results are reported in Table VI, where CDA indicates the model trained by using CDA techniques (e.g., random

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TABLE VI
PERFORMANCE COMPARISON BETWEEN BMDA AND CDA TECHNIQUES

DOTA				
Method	CDA	BMDA	mAP	Gain
SRAF-Net			61.64	
	✓		62.03	+ 0.39
		✓	64.99	+ 3.35
	✓	✓	**65.73**	**+ 4.09**

flip/crop/rotate/scale/expand). The comparison demonstrates that our BMDA is more effective than the aforementioned data augmentation techniques. By only applying with BMDA, the SRAF-Net can obtain a significant improvement of 3.35% mAP. The BMDA data augmentation technique can improve the robustness of the detector by images and labels mixup at the training stage.

V. CONCLUSION

We propose an effective detector, namely, SRAF-Net, for object detection in RSIs. The proposed SRAF-Net is an end-to-end training approach to integrate both scene-contextual information and multiscale capabilities into a deep convolutional neural network. It can realize a solution to the design difficulties of anchor-based detectors in RSIs and lack scene-level contextual information. In our SRAF-Net, we design an effective FPN for extracting multiscale scene-level contextual features. Moreover, a novel SADH is presented by integrating global scene- and local instance-level object classification scores for more accurate detection. The superiority of the proposed SRAF-Net has been demonstrated by comparison against the SOTA anchor-based and anchor-free detectors on the standard benchmarks. In the future, we would like to simplify the model by using a more simple feature extraction framework and apply the idea to many detectors since the designed modules, such as SADH and SE-FPN, are general.

ACKNOWLEDGMENT

The authors would like to thank Zhi Tian from The University of Adelaide for sharing his code for FCOS available at https://github.com/VectXmy/FCOS.Pytorch. They also thank the editors and three reviewers for their valuable comments and suggestions, which led to a substantial improvement of this article.

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